ALGEBRA 1 · PERSONALIZED PRACTICE

Math Practice

2026-11-01 · 20 questions

I CAN USE INPUT-OUTPUT RULES TO REPRESENT RELATIONSHIPS.

1. I can use input-output rules to represent relationships.

Write a rule.

Inputs 0,1,2 give outputs 4,7,10.

2. I can use input-output rules to represent relationships.

A function machine triples the input and subtracts 2. What output comes from input 5?

3. I can use input-output rules to represent relationships.

Input	-1	0	2
Output	1	3	7

Describe an operation rule that works for every pair.

4. I can use input-output rules to represent relationships.

A machine adds 4 to an input, then squares the result. Explain why $(x + 4)^2$ matches but $x^2 + 4$ does not.

I CAN DISTINGUISH FUNCTIONS FROM RELATIONSHIPS THAT ARE NOT FUNCTIONS.

5. I can distinguish functions from relationships that are not functions.

Identify the repeated input.

x	-1	2	-1
y	3	5	8

6. I can distinguish functions from relationships that are not functions.

For $y = 2x - 5$, find the output when $x = -1$.

7. I can distinguish functions from relationships that are not functions.

The pairs are $(1, 4), (2, 6), (1, 7)$. Is this relation a function? Explain.

8. I can distinguish functions from relationships that are not functions.

Function or not?

$(0, 1), (2, 3), (0, 4)$

I CAN IDENTIFY THE DOMAIN AND RANGE OF A RELATION.

9. I can identify the domain and range of a relation.

List the possible inputs.

A parking garage has levels 1 through 5. List the possible level inputs.

10. I can identify the domain and range of a relation.

List the range.

$(2, 5), (4, 7), (6, 5)$

11. I can identify the domain and range of a relation.

Tickets cost $6 each, and at most 8 tickets may be bought. What inputs are reasonable for the total-cost rule?

12. I can identify the domain and range of a relation.

x	-2	0	2	4
y	5	1	5	9

List the domain and range without repetitions.

I CAN CONNECT LINEAR GROWTH IN TABLES, GRAPHS, AND EQUATIONS.

13. I can connect linear growth in tables, graphs, and equations.

Find the change per step for outputs 5, 9, 13, 17. Is $(1, 2), (2, 2), (1, 3)$ a function?

14. I can connect linear growth in tables, graphs, and equations.

Complete the table.

x	0	1	2	3
y	4	7	10	?

15. I can connect linear growth in tables, graphs, and equations.

Find the slope.

Points (1,3) and (5,11)

16. I can connect linear growth in tables, graphs, and equations.

A table has $x = 0, 1, 2, 3$ and $y = 3, 7, 11, 15$. Write a rule.

I CAN CALCULATE AND INTERPRET SLOPE AS A CONSTANT RATE OF CHANGE.

17. I can calculate and interpret slope as a constant rate of change.

Find the slope through $(0, 2)$ and $(3, 8)$.

18. I can calculate and interpret slope as a constant rate of change.

A tank has 12 L at minute 1 and 27 L at minute 6. Find and interpret its average rate of change.

19. I can calculate and interpret slope as a constant rate of change.

Find the rise.

$(3,-4)$ to $(7,5)$

20. I can calculate and interpret slope as a constant rate of change.

Find the slope.

$(-2,1)$, $(2,9)$